F70LB Life Insurance Mathematics B

Heriot-Watt University

2025-2026

2 Policy values and reserves

In this section, we will look at policy values and reserves. Policy values are a way for the life office to estimate how much money it needs now to be able to cover its net future payments. In other words, policy values are a way to calculate (the value of) liabilities.

This is important for the life office's balance sheet, so in this section we will start there. You have already met policy values in F70LA Life Insurance Mathematics A last semester and we will briefly recap some of those details. We also explain how to calculate policy values for with-profit policies. We introduce the future loss random variable, which underlies our idea of policy values.

We also recap recursive relationships between policy values. Finally, we discuss asset shares and the connection with retrospective and prospective policy values.

2.1 Putting reserves in context: The balance sheet

Let's start with a look a life office's balance sheet.

Every enterprise, life offices included, needs a balance sheet showing the values of its assets and its liabilities. Broadly speaking, if the value of the assets exceeds the value of the liabilities, the life office is solvent. Otherwise it is insolvent.

So what are the assets and liabilities of a life office?

- The majority of the **assets** will be investments such as bonds, equities and property. They have been purchased with policyholders' premiums and they will be held in a fund from which benefits will be paid out.
- The **liabilities** are the promises made to pay benefits in future, offset against policyholders' promises to pay premiums in future.

We are interested in the *value* of the assets and liabilities, i.e. a number that represents how much each is worth.

Most assets can be valued at market value (e.g. bonds and equities typically have a market value). A harder question is, how do we value the liabilities?

We can do this by calculating a *policy value*. The actuary makes estimates of future interest rates, mortality and possibly expenses. (This collection of assumptions is called a *valuation basis*.) Using these assumptions, the actuary estimates the assets required to meet the expected future payments to policyholders, net of future income. The resulting number is called a *policy value*. In other words, the policy value $V(t)$ at time t is defined to be:

$V(t) = EPV(\text{Outgo on the policy from time } t \text{ onwards } \textit{minus} \text{ income on the policy from time } t \text{ onwards})$

What outgo and income we allow for depends on the type of policy value we calculate, either a *gross premium policy value* or a *net premium policy value*. We come back to this in section 2.2. We give a more formal definition in section 2.2.3.

The total of all the policy values is then one way to calculate the liability shown in the balance sheet.

The reserve is then the portfolio of assets held by an insurer to ensure that it can pay its future liabilities. The reserve needs to be at least equal to the total of policy values.

An insurer needs to hold reserves for life insurance policies because the premium income does not happen at the same time as the benefits and expenses outgo. The outgo usually increases with duration since the start of the policy, but the premium income is usually level. In other words, we need to keep some of the premiums we receive at the start (not just pay them out to shareholders as dividends!) so we can use them to pay benefits later on.

So the reserve is not the same as a policy value. However, in this course we will often just assume they are the same.

Notice that policy values and reserves concern the **future**. We look at what outgo and income we have promised and make sure we expect to have sufficent reserves to cover the difference. Assets, on the other hand, arise from what has happened in the **past**. They reflect the accumulation of all the past income and outgo. In this way, the balance sheet summarises our position at a point in time.

2.1.1 A simple policy value example

As an example to show that income is earlier than expenditure on average, consider a whole life policy for a life aged 30 with sum assured £100k paid at the end of the year of death, interest of $i = 4\%$ and mortality in line with the AM92 table.

1. Calculate the level annual premium paid in advance.
2. Calculate the expected amount of benefits paid out each year
3. Create a graph to compare them
4. Consider what this means for reserves (or policy values)

In this simple example, the level annual premium can be calculated by hand using your Orange tables as

$$P = \frac{100000 A_{30}}{\ddot{a}_{30}} = £733.85$$

or you can use a spreadsheet. The expected claims paid at age $x + t$ is $100000 q_{x+t}$ which looks like this when plotted on a chart in Excel:

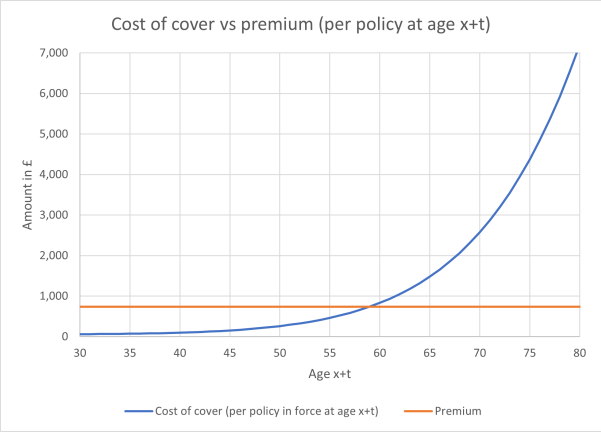


Figure 1: Premiums are level but expected claims are increasing

What this means for reserves: We can see that until almost age 60, we expect to receive more premium income per policy in force than we pay out in claims, i.e. the orange line is above the blue line. After this, it swaps. So before age 60, we get in more money than we pay out. We can't just give this to our shareholders (if we have them) because we need to be ready for the later years when we pay out more than we receive. This means that we need to keep a positive reserve.

2.2 A recap of policy values

In the previous section, we explained how policy values and reserves fit into the life office's business. Now, let's remind ourselves how policy values may be calculated. In this section, we will always consider the *prospective* (i.e., forward-looking) policy value unless otherwise stated. (We explain the connection with *retrospective* (i.e., backward-looking) policy values later.)

2.2.1 Gross premium policy values

The *gross premium* (or *office premium*) means whatever premium is actually paid by the policyholder. There are many different ways in practice that an insurer can choose what premium to charge. Fro example, it might have been calculated using a *premium basis* (i.e., set of assumptions) which is different to the valaution basis. The *gross premium policy value* for a policy at time t is calculated as follows:

$$V(t) = EPV(\text{future benefits} + \text{future expenses} - \text{future gross premiums})$$

For with-profit policies, in the gross premium policy value the future benefits include the following:

- declared bonuses
- an allowance for further bonuses being declared in future

With-profit policy values calculated in this way are called **bonus reserve** policy values.

2.2.2 Net premium policy values

Net premium policy values are perhaps easier to calculate, but less easy to understand because they are more artificial. Instead of using the *gross premium*, we must first calculate the *net premium*. Then the policy value is calculated as:

$$V(t) = EPV(\text{future benefits} - \text{future net premiums})$$

Note that there is no allowance for expenses. The *net premium* is the premium calculated using the policy value basis and the *principle of equivalence* allowing for the same benefits as the policy value. So the calculation of the *net premium* excludes expenses.

For with-profit policies, in the *net premium policy value* the future benefits include declared bonuses, but there is no allowance for any further future bonuses.

2.2.3 The loss random variable, L_t, and its mean

We now look at how to define policy values in terms of a random variable, starting from the example of a whole life policy. For a whole life policy issued to (x) with sum assured £1 and level annual premiums, the annual premium payable given some mortality and interest assumptions (and ignoring expenses), is:

$$P_x = \frac{A_x}{\ddot{a}_x}.$$

In effect we determined the premium that makes:

$$\text{EPV[benefits]} = \text{EPV[premiums]}$$

This is equivalent to:

$$\begin{aligned} & \mathrm{E}\left[v^{1+K_x}\right] = \mathrm{E}\left[P_x\, \ddot{a}_{\overline{1+K_x|}}\right] \\ \Rightarrow \quad & \mathrm{E}\left[v^{1+K_x}\right] - \mathrm{E}\left[P_x\, \ddot{a}_{\overline{1+K_x|}}\right] = 0 \\ \Rightarrow \quad & \mathrm{E}\left[v^{1+K_x} - P_x\, \ddot{a}_{\overline{1+K_x|}}\right] = 0. \end{aligned}$$

Definition. *At any future time t, we define the random variable L_t, called the loss at time t, to be the difference between the present values, at time t, of future outgo and of future income:*

$$L_t = PV[\text{Future Outgo} - \text{Future Income}].$$

Notice that L_t is a random variable, since it is the present value of the future random cashflows. The expected value of this random variable is the *(prospective) policy value* of the contract at time t, and is denoted $V(t)$.

As a specific example, the loss just before a premium payment date t is:

$$L_t = v^{1+K_{x+t}} - P_x \ddot{a}_{\overline{1+K_{x+t}|}}$$

Therefore the prospective policy value at that time is:

$$V(t) = \mathrm{E}\left[L_t\right] = A_{x+t} - P_x \ddot{a}_{x+t}.$$

Introducing the future loss random variable L_t has given us a slightly more formal way to look at policy values.

2.2.4 Some points to note about policy values

Note the following points about policy values:

1. The policy value above was calculated just **before** the premium then due, at an integer duration. It turns out that policy values of this kind make the formal mathematics as simple as it is possible to make it, so it is often assumed that policy values are calculated just before premium due dates when working out the mathematics.

   It is important to realise, however, that in practice a life office will usually have to calculate the policy values for all its in-force business on a fixed calendar date. Only by coincidence will this be an integer policy duration, for any randomly chosen policy.

2. Applications of policy values and reserves:

   - Reserves are needed to pay surrender values to policyholders who surrender, if the policy terms allow the payment of a surrender benefit.
   - Policy values and reserves are necessary for calculations related to policy alterations and conversions.

     For example, a policy-holder who has held a whole life policy for 8 years may request to change the cover to that of an endowment assurance, if the policy terms allow.
   - Policy values and reserves are important for demonstrating solvency.

     On at least one fixed day each year life offices are required to show that they are solvent by demonstrating that they have sufficient assets to set aside reserves at least equal to the total of the policy values of their in-force business.
   - For with-profit policies, policy values and reserves are used in determining bonus levels.
   - They are also used by proprietary companies as part of the determination of dividends to be paid to shareholders.

3. Notice that if the premium P is calculated using the principle of equivalence and the same basis as the policy value, then

   $V(0) = EPV(\text{Outgo on the policy from time 0 onwards } \textit{minus} \text{ income on the policy from time 0 onwards}) = 0$

4. Boundary conditions for policy values: When we calculate policy values, we often need to start from a boundary condition and proceed iteratively. There are often simple and obvious boundary conditions for policy values at outset or at the expiry of a policy.

- If the basis used to calculate the policy values and that used to calculate the premium are the same, then:

$$V(0) = 0$$

  This will always be the case for net premium policy value calculations.

- If a policy does not pay a benefit at maturity (e.g. an n-year term assurance policy), then:

$$V(n) = 0$$

- For a policy paying out a maturity benefit of S after the expiry at time n, then:

$$V(n) = S$$

5. The standard actuarial notation for various kinds of policy may also be extended to policy values using the symbol ${}_tV$, where t is the duration, and appending the usual subscripts and superscripts. For example:

Policy	Premiums	Policy Value
Whole-life	Annual	${}_tV_x$
Whole-life	m-thly	${}_tV_x^{(m)}$
n-year endowment	Annual	${}_tV_{x:\overline{n}\rceil}$

   and so on. However, we will normally just use the simpler notation $V(t)$.

2.2.5 Recursive relationships between policy values

You met recursive relationships between policy values last semester. They are relevant where cashflows occur in discrete time, typically annually.

Consider a policy issued to a life (x) where cash flows (premiums, expenses and claims) can occur only at the start or end of a year. Suppose this policy has been in force for t years, where t is a non-negative integer. Consider the year $t \rightarrow t+1$ and let:

- P_t denote the premium payable at time t,
- e_t denote the premium-related expense payable at time t,
- S_{t+1} denote the sum insured payable at time $t+1$ if the policyholder dies in the year,
- E_{t+1} denote the expense of paying the sum insured at time $t+1$,
- $V(t)$ denote the policy value for a policy in force at time t, and,
- $V(t+1)$ denote the policy value for a policy in force at time $t+1$.

Let q_{x+t} denote the probability that the policy holder, alive at time t, dies in the year and let i_t denote the rate of interest assumed earned in the year. The quantities e_t, E_t, q_{x+t} and i_t are all as assumed in the policy value basis.

Then:

$$(V(t) + P_t - e_t)(1 + i_t) = q_{x+t}(S_{t+1} + E_{t+1}) + p_{x+t}\, V(t+1)$$

Recursive relationships are important for the following reasons:

1. Recursive relationships can be interpreted as statements about the evolution of the life office's balance sheet, thus: If the office invests in assets that earn the rate of return assumed in the basis, and mortality and (possibly) expenses are also exactly as in the basis, then at all times the office will hold assets exactly equal in value to the policy value, i.e. the liability.

2. These recursive relationships are valid for all forms of benefit, even benefits that may depend on the reserve.

3. They define the policy values at non-integer durations. You can practice examples of this in the tutorial sheets and in past exam papers.

This semester, we will expect you to be able to use these relationships in Excel (and on paper) to calculate policy values at integer or non-integer times. You may also be expected to prove that a recursive relationship holds. For example, the special case of the recursive relationship for a whole life policy is $(V(t) + P_x)(1 + i) = q_{x+t} + p_{x+t}\, V(t + 1)$. It can be proved as follows:

$$\begin{aligned}(V(t) + P_x)(1 + i) &= (A_{x+t} - P_x\, \ddot{a}_{x+t} + P_x)\,(1 + i)\\ &= [A_{x+t} - P_x(\ddot{a}_{x+t} - 1)]\,(1 + i)\\ &= [A_{x+t} - P_x(\, a_{x+t})]\,(1 + i)\\ &= \frac{vq_{x+t} + vp_{x+t}A_{x+t+1} - P_x(vp_{x+t}\ddot{a}_{x+t+1})}{v}\\ &= q_{x+t} + p_{x+t}\,(A_{x+t+1} - P_x\, \ddot{a}_{x+t+1})\\ &= q_{x+t} + p_{x+t}\, V(t + 1).\end{aligned}$$

Example: Consider a 30-year endowment sold to a life age 30, with death benefit payable at the end of the year of death. Using AM92 mortality and 4% interest, evaluate:

1. $V(4.5)$, assuming premiums are payable annually, given that $V(5) = 0.097404$
2. $V(14.75)$, assuming premiums are payable quarterly.

Solution:
(i) No premium is paid between $t = 5$ and $t = 4.5$, hence:

$$V(4.5)(1 + i)^{0.5} = {}_{0.5}q_{34.5} + {}_{0.5}p_{34.5}\, V(5).$$

How do we calculate ${}_{0.5}p_{34.5}$?
If we assume a uniform distribution of deaths between integer ages, then:

$${}_{0.5}p_{34.5} = 1 - \frac{0.5q_{34}}{1 - 0.5q_{34}}$$

From the tables, this gives us:

$${}_{0.5}p_{34.5} \approx 0.99967$$

and substitution gives us:

$$V(4.5) \approx 0.095805$$

(ii) Premiums are paid quarterly. Let $P^{(4)}_{30:\overline{30}|}$ be the annual amount of premium, then:

$$P^{(4)}_{30:\overline{30}|} = \frac{A_{30:\overline{30}|}}{\ddot{a}^{(4)}_{30:\overline{30}|}} = 0.018129$$

and:

$$V(15) = A_{45:\overline{15}|} - P^{(4)}_{30:\overline{30}|}\, \ddot{a}^{(4)}_{45:\overline{15}|} = 0.35704$$

Now the recursive relationship is:

$$\left(V(14.75) + 0.25\, P^{(4)}_{30:\overline{30}|}\right)(1 + i)^{0.25} = {}_{0.25}q_{44.75} + {}_{0.25}p_{44.75}\, V(15).$$

If we assume a uniform distribution of deaths between exact ages 44 and 45 we can calculate ${}_{0.25}p_{44.75}$ as:

$${}_{0.255}p_{44.75} = 1 - \frac{0.25q_{44}}{1 - 0.75q_{44}} = 0.99967$$

Then substitution gives us:

$$V(14.75) = 0.3263$$

2.3 Asset shares and retrospective policy values

The asset share at duration t represents the money the insurance company has at time t in respect of a policy still in force. It is relevant mainly to with-profit policies and is taken into account when determining the level of bonus to award.

Definition. *The asset share is the accumulation to time t of past premiums received minus past claims and past expenses paid. The asset share is calculated using actual experience of premiums, claims, expenses and asset returns.*

The asset share is calculated assuming the policy being considered is one of a large number, N, of identical policies issued simultaneously:

$$\text{Asset share} \;=\; \text{total accumulated funds/number of survivors}$$

(which does not depend on N).

Comparing the asset share and the policy value tells us whether we have accumulated sufficient assets in respect of that policy to cover the expected future net cashflows on the policy. The following result gives a situation in which the asset share and policy value will be exactly the same:

Result. *Suppose the following three conditions all hold:*

- *the premium is calculated using the equivalence principle,*
- *the premium basis is the same as the policy value basis, and*
- *the past experience of the group of policies (survival, interest, expenses, bonus) is as assumed in this common basis*

Then Asset share = Policy value

In practice, this will not happen. The asset share is based on what has actually happened since the start of the policy, not the assumptions from the policy value basis. Any actuary will tell you that one thing we always know for sure about assumptions is that, however sensible the assumptions might be, experience will turn out to be at least a little bit different. The policy value will not usually turn out to be exactly the same as the asset share.

However, the result above does help us to understand the relationship between asset shares and policy values. In this course, we normally consider only the *prospective* policy value. However, policy values can also be defined retrospectively as follows :

Definition. *The retrospective policy value is the accumulation to time t of past premiums received minus past claims and past expenses paid. The retrospective policy value is calculated using the valuation basis, i.e. a set of assumptions.*

It turns out that the retrospective policy value and the prospective policy value are the same if they and the premiums are all calculated on the same basis. This will hold for net premium policy value calculations, in particular, which can make the retrospective policy value a helpful calculation tool for net premium.

By comparing the two definitions above, you can see that the mathematical calculation of the asset share and the retrospective policy value are the same except for the parameters used. The asset share reflects what really happened, but the retrospective policy value uses assumptions.

The retrospective policy value is interesting to help us understand the mathematical connection between the prospective policy value and the asset share, but it is the prospective policy value that is mainly of interest. The prospective policy value tells us about what reserves we need.